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SYSTEMATIC REVIEW

The Relationship Between Training Load and Injury in Athletes: A Systematic Review

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Abstract

Background The relationship between training load and musculoskeletal injury is a rapidly advancing area of research in need of an updated systematic review.

Objective This systematic review examined the evidence for the relationship between training load and musculoskeletal injury risk in athlete, military, and first responder (i.e. law enforcement, firefighting, rescue service) populations.

Methods The CINAHL, EMBASE, MEDLINE, SportDISCUS, and SCOPUS databases were searched using a comprehensive strategy. Studies published prior to July 2017 were included if they prospectively examined the relationship between training load and injury risk. Study quality was assessed using the Newcastle–Ottawa Quality Assessment Scale (NOS) and Oxford Centre for Evidence-Based Medicine levels of evidence. A narrative synthesis of findings was conducted.

Results A total of 2047 articles were examined for potential inclusion. Forty-six met the inclusion criteria and 11 known to the authors but not found in the search were added, for a total of 57 articles. Overall, 47 studies had at least partially statistically significant results, demonstrating a relationship between training load and injury risk. Included articles were rated as poor ($n=15$), fair ($n=6$), and good ($n=36$) based on NOS score. Articles assessed as 'good' were considered level 2b evidence on the Oxford Centre for Evidence-Based Medicine Model, and articles assessed as 'fair' or 'poor' were considered level 4 evidence.

Conclusions Our results demonstrate that the existence of a relationship between training load and injury continues to be well supported in the literature and is strongest for subjective internal training load. The directionality of this relationship appears to depend on the type and timeframe of load measured.

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Key Points

The relationship between training load and injury has become firmly established in the literature due to a large number of recent well-designed studies.

The strongest evidence is for internal training load assessed using session rating of perceived exertion, and relative training loads assessed using the acute:chronic workload ratio.

The evidence for a relationship between training load in the form of session frequency and injury is conflicting.

There is some evidence that a training load–injury relationship exists in military populations, although the number of studies is limited.

The relationship between training load and injury has variously been described as direct, inverse, and U-shaped, depending on the timing and measure of load used.

1 Introduction

Prediction of musculoskeletal injury risk in athletes is an area of increasing research activity due to rising levels of competition and the impact of player availability on success in elite sports organizations [1, 2]. Prediction of musculoskeletal injuries in occupational settings, particularly in military personnel and other 'tactical athletes' has also been a recent focus [3–5] due to the detrimental impact training injuries have on the readiness of these service personnel [6]. A significant challenge in predicting injury risk is the multifactorial nature of musculoskeletal injuries [7]. A large body of evidence has identified several risk factors associated with injury risk, but many of these are non-modifiable, such as age [8], sex [8], injury history [9–12], sport [13], and level of competition [8]. While knowledge of non-modifiable risk factors provides essential information for multivariable injury prediction modeling, these risk factors provide minimal avenues for sports medicine professionals to prevent injuries. Risk factors that have been shown in the literature to be both modifiable and prospectively associated with injury incidence are more useful elements in injury prediction modeling because they provide clinicians with intervention points that can be used to reduce the risk of injury.

Training load is a modifiable risk factor that has been examined in many recent studies and found to be prospectively associated with injury risk in athletes, making it potentially useful for injury prediction and prevention. Training load is defined as the cumulative amount of stress placed on an individual from single or multiple training sessions over a period of time [1, 14]. It can be measured along several dimensions, including external/internal, subjective/objective, and absolute/relative [1]. An external training load (ETL) is defined as any external stimulus applied to an athlete independent of their internal characteristics [15]. Examples of ETL include distance, acceleration, matches played, and pitches thrown [1]. Internal training load (ITL) involves quantifying an athlete's response to an external load [14]. Measures of ITL can be subjective or objective [16]. Subjective measures are those that are self-reported, such as session/s rating of perceived exertion (sRPE) [17] or questionnaires on well-being [16], while objective measures of ITL are those that are not self-reported, such as heart rate (HR) and blood lactate [16]. Only measures of ITL can vary in the subjective/objective dimension. The third dimension of load quantification is absolute/relative. Absolute loads are the summation of load values over a given period. Relative loads are those that take into account "the rate of load application, history of loading, or fitness level of the athlete" [1], and most often express variations in load between two time periods. Examples include the difference in load between two successive weeks or the ratio of a recent time period (commonly the prior 7 days) to that of a preceding time period (commonly the prior 28 days), termed the acute:chronic workload ratio (ACWR) [18]. The terms 'training load', 'workload', and 'load' are often used interchangeably in the literature, and all refer to the same construct. Hereafter, the term 'load' will be used to refer to this construct.

Early rapid growth in load–injury research permitted two early systematic reviews to be published on the topic (Jones et al. [19] and Drew and Finch [20]), both using literature searches completed in 2015. These early reviews both concluded that there was 'emerging moderate evidence' for a relationship between training load and injury risk in athletes [19, 20]. Jones et al. also noted a lack of consistency of definitions pertaining to training load in the literature reviewed [19]. Recently, there have been three significant changes in the literature that pertain to the completion of a systematic review on this topic. First, several new articles have been published that examine the relationship between training load and injury across several sports. Second, definitions of training load and related key terms were formalized in a consensus statement by the International Olympic Committee (IOC), complete with an extensive dictionary and recommendations for use of these terms in practice and research [1]. This consensus statement included measures of session frequency in its designations of training load measures, a point of inconsistency in the reviews by Jones et al. [19] and Drew and Finch [20], making its inclusion in subsequent reviews essential. Third, recent years have seen an increase in the training and recognition of military personnel and

first responders (law enforcement officers, firefighters, and rescue service personnel) [21] as tactical athletes [21–24]. Similar to traditional, or competitive, athletes, tactical athletes participate in rigorous [22] and periodized [25] training programs with the goals of optimizing performance and preventing injuries [26]. These three developments necessitate the completion of an updated systematic review on this topic. Thus, the purpose of this systematic review was to update and expand upon the reviews conducted by Drew and Finch and Jones et al. by examining the evidence for the relationship between training load and musculoskeletal injury risk (1) in light of the many studies on the topic completed since publication of these reviews, (2) by extending the definition of training load to include session frequency in accordance with the IOC consensus statement, and (3) by extending the review to include studies conducted in tactical athletes in addition to competitive athletes. Secondary purposes of this review were to investigate if sufficient evidence exists in terms of quantity and quality of studies to determine (1) if measures of ITL are stronger predictors of injury than measures of ETL, and (2) if subjective measures of ITL are stronger predictors of injury than objective measures of ITL.

2 Methods

2.1 Search Strategy

This systematic review was registered in the PROSPERO database (CRD42017058881) and conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [27]. The research question and inclusion criteria were established a priori. Two deviations from the inclusion criteria were made during the course of the study. First, studies examining athletes under the age of 18 years were included because we determined there was insufficient evidence that the load–injury relationship differs between youth and adult athletes. Second, the requirement that a measure of load be reported in the form of a second-order unit (a product, quotient, ratio, or percent) was removed due to the determination that traditional measures of training volume, consisting of simple sums, such as weekly running mileage, qualified as a measure of load under the definition standardized in the IOC consensus statement. Both of these changes were made early in the screening phase, when fewer than 200 of the 2047 total studies had been screened.

Pertinent studies were obtained by searching the CINAHL, EMBASE, MEDLINE, SportDISCUS, and SCOPUS databases. Search terms and Medical Subject Headings (MeSH) used in the search were developed in collaboration with a librarian-scientist who specialized in conducting health science systematic reviews. Searches included terminology relevant to training load monitoring and musculoskeletal injury in athletes, military personnel, and first responders. The full search strategy is available in electronic supplementary material Appendix S1. Search limitations included original research articles published in peer-reviewed journals in the English language. No limits were placed on year of publication. Articles obtained were organized using EndNote™ X7.7.1 software (Thomson Reuters, 2016), and the total number of articles obtained from each database was recorded before and after removal of duplicates. After removal of duplicates, the titles and abstracts of all articles were screened independently by two reviewers (TE and DH) using Covidence™ software (Covidence, 2017) accessed online at www.covidence.org. Articles were placed into separate 'yes', 'no', and 'maybe' folders based on the stated inclusion/exclusion criteria. All articles in the 'maybe' folder were reviewed by a third reviewer (DP) for possible inclusion. The two reviewers then compared 'yes' folders for agreement. An attempt to resolve any disagreements was made, with the third reviewer settling any remaining disagreements as needed. Full-text versions of all studies in the 'yes' folder were then obtained and rechecked against the inclusion/exclusion criteria independently by two reviewers (TE and DH) for final inclusion. A second search was conducted after the completion of the initial data extraction and synthesis phases. Additional studies identified in this search that met the inclusion criteria were included in the extraction and synthesis described above. The second search was conducted on 24 July 2017. Studies known to the authors published on or before 1 July 2017 that met the inclusion criteria but were not found in the search were also included.

2.2 Study Selection

Studies that met the following criteria were included. Studies could be of any design, with the exception of case studies, case series, and case–control studies, and were limited to those prospective in nature, i.e. only cohorts followed into the future, as defined by the Cochrane Collaboration [28], for collection of both training load and injury data. Conference proceedings and abstracts were not included. Study participants were limited to human athletes (amateur or professional), military personnel, and first-responders, with no limitations placed on age or sex. Studies were limited to those that calculated at least one measure of training load, measured as external or internal, subjective or objective, absolute or relative, taken during practice, competition, or physical training. To qualify as a measure of training load, a measure was required to have a component of session quantity (i.e. time, distance, number of repetitions), session intensity, or both. Studies were also limited to those that reported musculoskeletal injuries or injury status as an outcome,

either self-reported or diagnosed by a clinician. Finally, studies were limited to those that reported some measure of the strength of association between the incidence of musculoskeletal injury and training load. Studies that used inferential statistics to test for significant differences in training load between injured and uninjured athletes or between groups with different injury incidence were also included, as were studies that used inferential statistics to test for significant differences in injury incidence between groups with different training loads. Studies that did not meet any one or more of these specific inclusion criteria were excluded.

2.3 Data Extraction

Data extracted from each article included year of publication, study design, study location, population, sample size, sample demographics (age, sex, occupation/sport/position), type of training load data collected, units in which training loads data were reported [e.g. km/week, throws/match arbitrary units [AU)], reported statistics for the main study results [odds ratios (OR), correlations, mean differences, injury rate ratios (IRR), injury proportion ratios (IPR), *t* tests, ANOVAs, Chi square tests), and their statistical significance. A standardized data collection form was used to extract all data.

2.4 Study Quality and Level of Evidence

The quality of each study, including the potential for bias, was assessed using the Newcastle–Ottawa Quality Assessment Scale (NOS) for cohort studies [29]. The NOS is a quality assessment tool for non-randomized study designs that assesses a study in three areas: participant selection, comparability, and outcome [29]. Along with the Downs and Black Checklist for Measuring Quality (D&B) [30], the NOS is one of two tools recommended by the Cochrane Collaboration to evaluate the methodological quality of non-randomized studies [31]. The NOS has been shown to have moderate to good interrater reliability [intraclass correlation coefficient (ICC) 0.52, confidence interval (CI) 0.14–0.76] when used in observational studies with musculoskeletal health outcomes [32]. Although this is not as good as the reliability reported for the D&B, the NOS provides good performance with greater ease of use and more specific criteria relating to observational studies compared with the D&B [29, 30]. For these reasons, use of the NOS was selected a priori. Each article was appraised by two authors independently and all discrepancies in scoring were resolved by arbitration between the two reviewers. In cases where discrepancies could not be resolved, a third reviewer assessed the item in question. No blinding to any aspect of the reviewed studies was used for reasons of practicality. The level of evidence for each study that met the inclusion criteria was assessed using the Oxford Centre for Evidence Based Medicine Model [33]. Copies of the NOS and Oxford Centre levels of evidence are available in electronic supplementary material Appendices S2 and S3, respectively. Using the Van Tulder et al. method, the overall level of evidence was assessed as strong, "consistent findings among multiple high-quality randomized controlled trials (RCTs)"; moderate, "consistent findings among multiple low-quality RCTs and/or non-randomized controlled trials (CCTs) and/or one high-quality RCT"; limited, "one low-quality RCT and/or CCTs, conflicting evidence"; conflicting, "inconsistent findings among multiple trials (RCTs and/or CCTs)"; and no evidence, "no RCTs or CCTs" [34].

2.5 Data Synthesis

A narrative synthesis of the data was completed by reporting the findings on the relationship between training load and musculoskeletal injury from each study. The number of articles using each type of training load was reported, along with comparisons within and between groups of studies using the same type of training load. Strength of evidence was evaluated based on the number of studies examining each type of load, the proportion with significant findings, and the methodological quality of the studies. Methodological quality (NOS) and level of evidence of each study were also reported separately for each study. Because there were an insufficient number of studies examining tactical athletes, no separate synthesis of these studies was performed.

3 Results

3.1 Article Identification

Our search strategy identified 2036 potential articles for inclusion, after removal of 1181 duplicates. After exclusion of articles that did not meet the inclusion criteria based on title and abstract review, 92 articles remained. Review of the full-text versions of these 92 articles resulted in the removal of 46 additional articles, leaving 46 remaining articles. Eleven additional articles meeting the inclusion criteria but not found in our search were also screened and included, resulting in a total of 57 articles included in our final dataset. A flowchart diagram summarizing the article identification process is shown in Fig. 1.

3.2 Descriptions of the Included Articles

All 57 of the included articles were cohort studies. The greatest number were from Australia ($n = 28$) and Europe ($n = 22$), with New Zealand ($n = 2$), North America ($n = 2$), and South America ($n = 1$) accounting for the rest. Two

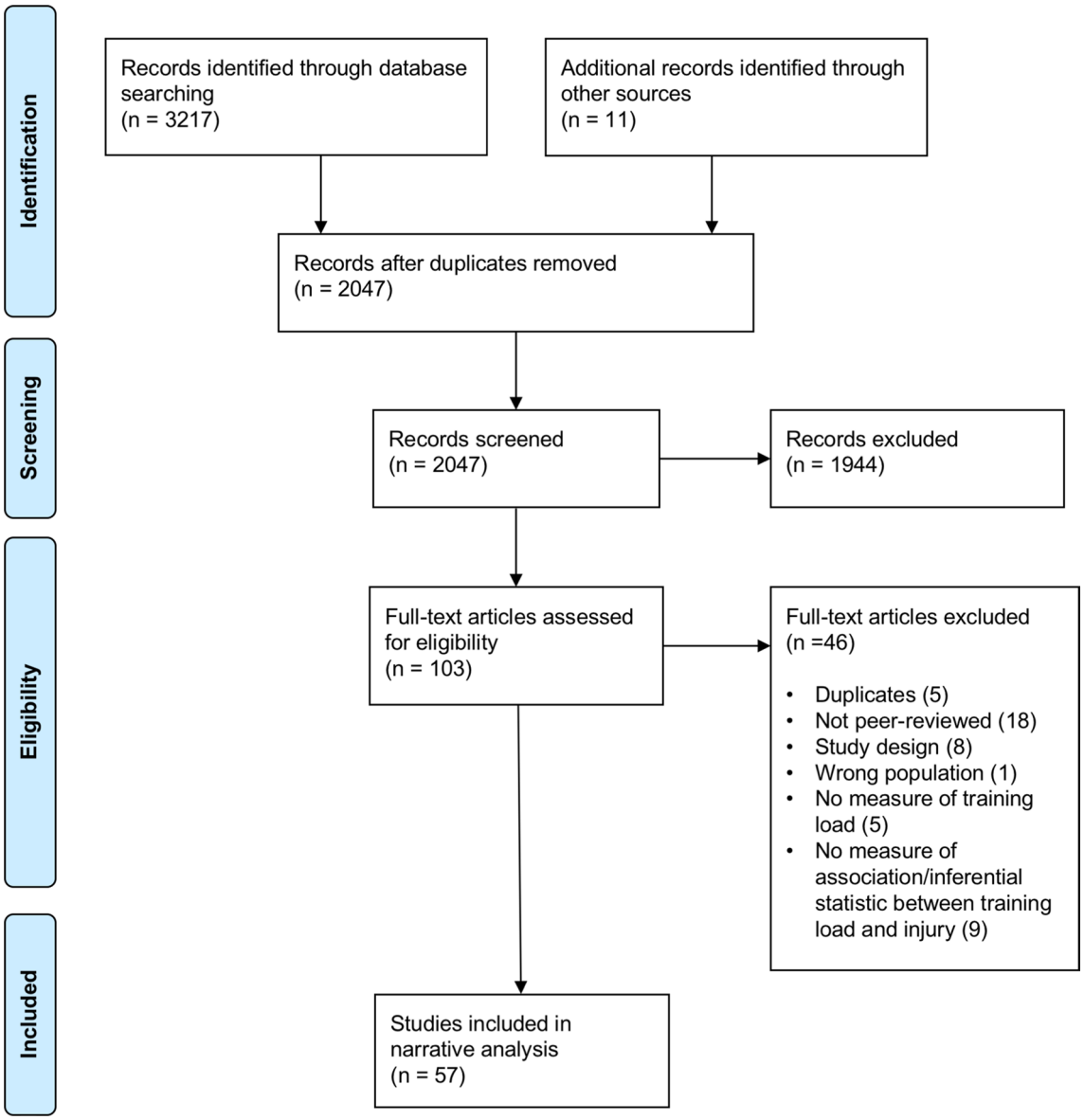

Fig. 1 Study selection process

articles did not specify location of origin. The populations most frequently studied were rugby ($n = 14$), soccer ($n = 13$), and Australian Rules Football ($n = 9$) athletes. Two included studies examined tactical athlete populations, both of which were conducted in military personnel. Most studies ($n = 44$) had all male participants, two had all female participants, and ten had both. The sex of participants in one study was not discernable. Six studies included youth athletes, with five of these exclusively examining youth athletes. Sample size ranged from 12 to 1696, with nine studies having 30 or fewer participants, 21 having 31–60 participants, eight having 61–100 participants, 12 having between 101 and 500 participants, and seven having more than 500 participants. Article characteristics are presented in Table 1.

3.3 Measures of Load and Findings

Thirty-two studies examined some form of ETL, 14 examined ITL, and 11 examined both. Overall, 47 studies had at least partially statistically significant results, demonstrating a relationship between training load and injury risk, with 32 studies reporting all statistically significant results. Conversely, 25 articles had at least partially null results, with ten studies reporting all null findings. Of the studies with statistically significant findings, 29 reported a direct relationship between load and injury, seven reported an inverse

Table 1 Summary of methods and results of the included studies

Study	Participants	Training load	Results
Ekstrand et al., 1983 [50] Prospective cohort study Sweden	$n = 180$ Senior-level soccer players All males Mean age 24.6 years (SD 4.6)	*ETL* (1) Session (practice and competition) frequency (2) Total training time *Units* (1) Practice:competition ratio (2) Total training hours	Teams with less-than-average training experienced significantly increased injury IR when training was increased ($p < 0.01$), and teams with higher-than-average training experienced decreased rates when training was increased ($p < 0.01$). However, statistical test here is unclear. A significant direct correlation was also found between injury incidence rate and the practice:game ratio, although this significance required the removal of the data of one outlier team. The statistical test used is unclear
Korkia et al., 1994 [76] Prospective cohort study UK	$n = 155$ Recreational, intermediate, or elite triathletes 124 males 31 females *Mean age* Males 34 years (SD 8.9) Females 32 years (SD 7.3)	ETL measured in four ways: (1) Number of sessions per week (2) Number of training hours per week (3) Total distance swimming, biking, and running (4) Minutes per week at hard, moderate, and easy intensities and performing fartlek and hill training *Units* (1) Number of sessions per week (2) Number of hours per week (3) Distance in km per week (4) Minutes per week	There were no statistically significant associations between average distance per week swimming, biking, or running and injury incidence. There was also no statistically significant relationship between the number of sessions or hours per week and injury incidence. In addition, no statistically significant association was found between intensity and injury incidence. What statistics were used is unclear
Hamalainen et al., 1994 [79] Prospective cohort study Finland	$n = 66$ Finnish Air Force Academy pilots All males Age range 20–28 years	*ETL* Time spent flying fighter planes *Units* Total hours spent flying	A two-sample t test found a greater number of flight hours in participants with vs. without neck pain ($p = 0.01$)
Anderson et al., 2003 [82] Prospective cohort study USA	$n = 12$ NCAA Division III collegiate basketball players All females Age range 18–22 years	*ITL* sRPE and duration in minutes for training and matches *Units* AU	A PPMCC was used to assess the association between weekly average team training load and number of injuries in the team each week ($r = 0.675$, $p < 0.01$)

Table 1 (continued)

Study	Participants	Training load	Results
Dennis et al., 2003 [45] Prospective cohort study Australia	$n = 90$ State-level cricket fast bowlers, cannot discern professional/amateur status All males Mean age 27 years (range 18–38)	ETL measured two ways (1) Average number of bowls delivered in matches, and training per week (only weeks in which the player bowled were included in the calculations) (2) Average number of days between matches *Units* (1) Average number of deliveries/week (2) Average number of days between matches	An independent *t* test found no difference in the mean number of deliveries/week between injured and uninjured bowlers ($p > 0.05$). A trend analysis was also used, which found that compared with those with an average of 123–188 deliveries per week, those with < 123 deliveries were at significantly greater risk of injury (RR 1.4, 95% CI 1.0–2.0). Those with > 188 deliveries were also at greater risk (RR 1.4, 95% CI 0.9–1.6) Injury risk between groups with differing average number of days between bowling sessions was also examined using a trend analysis. Average number of days were divided in groups ranging from < 2 to ≥ 5 days, and compared to the reference group of 3.0–3.99 days between sessions. Those with < 2 days were significantly more likely to be injured (RR 2.4, 95% CI 1.6–3.5), as were those with ≥ 5 days between sessions (RR 1.8, 95% CI 1.1–2.9)
Taunton et al., 2003 [41] Prospective cohort study Canada	$n = 844$ Recreational runners 205 males 635 females *Age* *Males* Under 30 years: 25 (12.3%) 31–49 years: 105 (51.5%) 50–55 years: 39 (9.1%) Over 56 years: 1 (< 1.0%) *Females* Under 30 years: 116 (18.6%) 31–49 years: 397 (63.6%) 50–55 years: 72 (11.5%) Over 56 years: 11 (< 1.0%)	*ETL* Frequency of running *Units* Days of running per week	Multivariate logistic regression examined running frequency in days per week along with other non-training load-related variables separately for each sex. This analysis demonstrated that, relative to the reference group of 3 days/week, women who ran 1 day/week were significantly more likely to be injured (OR 3.648, 95% CI 1.082–12.297)
Dennis et al., 2004 [51] Prospective cohort study Australia	$n = 12$ State-level cricket fast bowlers All males Mean age 25.2 years (range 21–34)	ETL measured two ways: (1) Number of days since previous session (2) Number of bowls delivered in matches and training per week, month, and season (excluding weeks in which the athlete did not bowl) *Units* (1) Average number of days since previous session (2) Average number of sessions/week and deliveries/week	Using ANOVA, injured bowlers had significantly fewer days between sessions ($p < 0.01$), no difference in number of deliveries per session (no *p*-value reported), but a significantly greater number of deliveries per week ($p < 0.01$) and per month ($p < 0.01$), as well as significantly more deliveries in the 8–21 days prior to injury ($p = 0.02$)

Table 1 (continued)

Study	Participants	Training load	Results
Gabbett and Jenkins, 2011 [88] Prospective cohort study Australia	$n = 79$ Semi-professional rugby players All males No age information	*ITL* sRPE and duration in minutes for training and matches *Units* AU	PPMCC were used to assess the strength of the relationship between changes in training intensity and training injury incidence ($r = 0.83$, $p < 0.05$), training duration and training injury incidence ($r = 0.73$, $p < 0.05$), and training load and training injury incidence ($r = 0.86$, $p < 0.05$). Match intensity and match injury incidence ($r = 0.74$, $p < 0.05$), match playing duration and match injury incidence ($r = 0.86$, $p < 0.05$), and match load and match injury incidence ($r = 0.86$, $p < 0.05$) were assessed the same way
Gabbett, 2004 [83] Prospective cohort study Australia	$n = 220$ Sub-elite (amateur) rugby players All males *Mean ages* Cohort 1: 22.9 years (20.7–25.1) Cohort 2: 19.6 years (18.4–20.8) Cohort 3: 21.5 years (19.6–23.4)	*ITL* sRPE and duration in minutes for training and matches *Units* AU	*Mean ITL* Cohort 1: 326 AU Cohort 2: 260.5 AU Cohort 3: 297.8 AU Differences in training load (Chi-square tests): Cohort 1 vs. 2: Yes ($p < 0.001$) Cohort 1 vs. 3: Yes ($p < 0.001$) Cohort 2 vs. 3: No ($p > 0.05$) Differences in injury IR (Chi-square tests): Cohort 1 vs. 2: Yes ($p < 0.001$) Cohort 1 vs. 3: Yes ($p < 0.001$) Cohort 2 vs. 3: No ($p > 0.05$)
Gabbett and Domrow, 2005 [38] Prospective cohort study Australia	$n = 153$ Professional rugby players All males Mean age 26 years (SD 4)	*ETL* Amount of training prior to first injury *Units* Number of weeks of training participation prior to first injury	A multivariate logistic regression equation using GEE adjusted for body mass, age, experience, grade, position, speed, muscular power, VO2max, matches played, severity of initial injury, body region of initial injury, and whether first injury was contact or non-contact demonstrated significantly greater risk for second injury in those players who participated within < 18 weeks prior to their first injury vs. those who participated ≥ 25 weeks (OR 8.69, 95% CI 1.33–56.76)
Gabbett and Domrow, 2007 [84] Prospective cohort study Australia	$n = 183$ Sub-elite (amateur) rugby players All males Mean age 21.4 years (SD 5.1)	*ITL* sRPE and duration in minutes for training sessions *Units* AU (duration assumed to be the same for all athletes)	*Individual model OR* Pre-season: 2.12 (95% CI 1.16–3.07, $p = 0.01$) Early-competition season: 2.85 (95% CI 1.41–4.28, $p = 0.01$) Late-competition season: 1.50 (95% CI 1.02–2.98, $p = 0.04$) Linear regression model coefficients for ITL for team-level data: Pre-season: 0.35 ($p = 0.01$) Early-competitive season: −0.08 ($p = 0.53$) Late-competitive season: 0.02 ($p = 0.84$)

Table 1 (continued)

Study	Participants	Training load	Results
Brooks et al., 2008 [85] Prospective cohort study UK	$n = 502$ Professional rugby players All males No age information provided	*ETL* Hours of training/match play *Units* Hours	A PPMCC demonstrated no association between injury incidence and hours of training per week (divided into quintiles) [p value not reported]
Gianoudis et al., 2008 [86] Prospective cohort study Australia	$n = 46$ Youth amateur basketball players on school teams 28 males 18 females Mean age 16.0 years	*ETL* Physical activity duration *Units* Hours of physical activity per week	An independent t test found no difference in the average number of hours of weekly physical activity between injured and uninjured athletes ($p = 0.67$)
Orchard et al., 2009 [53] Prospective cohort study Australia	$n = 129$ Professional cricket fast bowlers All males No age information provided	*ETL* Number of overs (a group of six deliveries) delivered *Units* (1) Number of overs delivered in the entire match (2) Number of overs delivered in each innings of the match (first and second) considered individually	A t test comparing the number of overs bowled per match was significantly higher for those who went on to incur an injury, with the greatest difference occurring in the period between 21 and 28 days ($p < 0.01$). RRs comparing the risk of injury in those who bowled > 50 vs. < 50 overs in a match demonstrated the greatest risk at 21 days (RR 1.77, 95% CI 1.05–2.98) and 28 days (RR 1.62, 95% CI 1.02–2.57)
Brink et al., 2010 [58] Prospective cohort study Denmark	$n = 53$ Elite youth soccer players All males Age range 15–18 years	*ETL and ITL* *ETL* Weekly hours of training and competition *Units* Time in hours *ITL* sRPE and duration in minutes for training sessions and competitions, although using Borg's original 15-point scale *Units* AU	A multinomial logistic regression model with traumatic injury, overuse injury, and no injury as potential outcomes found traumatic injury was related to both ETL (OR 1.14, 95% CI 1.06–1.23) and ITL (OR 1.01, 95% CI 1.00–1.02) in the previous week. There was no association between overuse injuries and either ETL (OR 1.07, 95% CI 0.98–1.18) or ITL (OR 1.01, 95% CI 1.00–1.02)
Dupont et al., 2010 [74] Prospective cohort study UK	$n = 32$ Professional soccer players All males Mean age 25.6 years (SD 3.8)	*ETL* Number of matches played per week *Units* Days between matches	A paired t test was used to compare injury IR between players during periods in which they participated in two matches (for at least 75 min each) within 4 days vs. those who had not participated in a match for 6 days or more. The injury IR was significantly greater during periods in which they participated in two matches (25.6 vs. 4.1 injuries per 1000 exposure hours, $p < 0.01$)

Table 1 (continued)

Study	Participants	Training load	Results
Killen et al., 2010 [68] Prospective cohort study Australia	$n = 36$ Professional rugby players All males Age range 17–32 years	*ITL* sRPE and duration in minutes for training and matches *Units* AU	PPMCCs were used to assess the association between weekly training load and weekly injury IR ($r = 0.023$, $p = 0.941$). The higher training loads in the first half of the pre-season corresponded with a higher injury rate [8.7 (SD 6 3.4) per 1000 training hours] in comparison to the second half of the pre-season [5.36 (SD 6.3) per 1000 training hours] ($p \geq 0.05$)
Wilson et al., 2010 [87] Prospective cohort study Ireland	$n = 20$ International-level amateur rowers 12 males 8 females Mean age 26.25 years (SD 4.18)	*ETL* Training time *Units* Training time in hours	PPMCC demonstrated no association between injury IR and total hours of training per week ($r = 0.543$, $p = 0.068$), but did find a significant relationship between number of weekly hours on the ergometer and injury IR ($r = 0.75$, $p = 0.01$), number of hours performing heavy weight training ($r = 0.66$, $p = 0.02$), and core training ($r = 0.68$, $p = 0.01$)
Gabbett and Jenkins, 2011 [88] Prospective cohort study Australia	$n = 79$ Professional rugby players All males Mean age 23.3 years (SD 3.8)	*ITL* sRPE and duration in minutes for training and matches *Units* AU	PPMCCs were used to assess the association between overall training load and overall injury incidence ($r = 0.82$, $p < 0.01$), non-contact field injury incidence ($r = 0.82$, $p < 0.01$), and contact field injury incidence ($r = 0.80$, $p < 0.01$). Relationships were also examined between field training load and overall injury field rate ($r = 0.68$, $p < 0.05$), non-contact field injury ($r = 0.65$, $p < 0.05$), and contact field injury ($r = 0.63$, $p < 0.05$). Also examined were associations between strength and power training loads and strength and power injuries ($r = 0.63$, $p < 0.05$), contact field injuries ($r = 0.75$, $p < 0.01$), and non-contact field injuries ($r = 0.82$, $p < 0.01$). The association between field training loads and incidence of strength and power injuries was also examined and found to be non-significant (no p value reported)

Table 1 (continued)

Study	Participants	Training load	Results
Gabbett and Ullah, 2012 [43] Prospective cohort study Australia	$n = 34$ Professional rugby players All males Age 23.6 years (SD 3.8)	*ETL* GPS was used to measure distance, speed, and position, and triaxial accelerometers were used to measure acceleration in training sessions *Units* Distance in metres covered (1) at various ranges of speed in m/s, and (2) while undergoing various ranges of acceleration in m/s/s	The frailty model (an extension of the Cox proportional regression model for recurrent events) was applied to calculate the relative risk of soft tissue lower extremity injury after adjusting for all other training data. This model showed the risk of NTL injury was significantly higher when very-high-intensity running exceeded 9 m per session (RR 2.7, 95% CI 1.2–6.5, $p < 0.05$), and that a reduction in time-loss injuries occurred when very-low-intensity movement exceeded 542 m per session (RR 0.4, 95% CI 0.2–0.9, $p < 0.05$)
Mallo and Dellal, 2012 [70] Prospective cohort study Spain	$n = 35$ Professional soccer players All males Mean age 21.4 years (SD 2.4)	ETL and ITL measured multiple ways *ETL* (1) Number of training/match sessions (2) Duration of training sessions *ITL* (1) Intensity using HR *Units* *ETL* (1) Number of training sessions/matches (2) Training session duration in minutes *ITL* (1) Mean HR as a percentage of each participant's HRmax	PPMCC demonstrated an association between injury IR of muscle strain injuries and session intensity (mean HR) ($r = 0.72$, $p < 0.05$). No significant relationships were found between frequency or duration and injury incidence
Dellal et al., 2015 [42] Prospective cohort study France	$n = 16$ International-level soccer players All males Mean age 24.3 years (SD 3.2)	*ETL* Match (session) frequency *Units* Number of matches in various time periods	Fisher's exact tests demonstrated that match injury IR were higher during congested periods (six games in 18 days) vs. those periods of less congestion ($p < 0.01$). Another Fisher's exact text demonstrated that training injuries were lower during the same high congestion period than during less-congested periods ($p < 0.01$). Overall injury rates did not differ between the congested and less congested periods (p value not reported)

Table 1 (continued)

Study	Participants	Training load	Results
Jacobsson et al., 2013 [69] Prospective cohort study Sweden	$n = 292$ Adult and youth track and field athletes ranked in the top 10 in Sweden 161 females, 131 males Mean age: adults 24 years (range 18–27), youth 17 years (no range reported)	*ITL* Modified version of sRPE, in which 2 = light, 3 = moderate, and 5 = hard, was used for intensity, and total weekly training time in minutes was used for duration *Units* Training load scores in AU (intensity in modified sRPE × weekly duration in minutes) were divided into four quartiles called a 'training load rank index'. This was done separately for athletes of each event type and athlete category (youth/adult and female/male)	Cox proportional hazards regression analyses demonstrated that athletes in the third quartile had greater risk of injury compared with those in the first (hazard ratio 1.79, 95% CI 1.54–2.78, $p < 0.05$), and that those in the fourth quartile also had a greater risk than those in the first quartile (HR 1.79, 95% CI 1.16–2.74, $p < 0.05$). In addition, log-rank tests demonstrated shorter time to first injury across increasing quartiles of ITL, controlling for age, sex, event type, and previous injury history ($p = 0.019$)
Rogalski et al., 2013 [57] Prospective cohort study Australia	$n = 46$ Semi-professional Australian rules Football players All males Mean age 22.2 years (SD 2.9)	*ITL* (1) Absolute: sRPE and duration in minutes for training and matches (2) Relative: week-to-week changes in workload *Units* (1) AU (2) ITL difference between 2 weeks in AU	Logistic regression was used to compare players in four ordinal categories of cumulative ITL. Those who performed > 1750 AU weekly were significantly more likely to sustain an injury compared with those who performed < 1250 AU (OR 2.44, 95% CI 1.28–4.66, $p = 0.007$). Those who performed 2-week loads > 4000 AU were significantly more likely to be injured compared with those whose 2-week loads were < 2000 AU (OR 4.74, 95% CI 1.14–19.76, $p = 0.033$). Players who experienced a previous to current week change in load of > 1250 AU during the season were also significantly more likely to experience an injury compared with those with a change < 250 AU (OR 2.58, 95% CI 1.43–4.66, $p = 0.002$). No significant ORs were calculated between groups in the pre-season
Theisen et al., 2013 [44] Prospective cohort study Luxembourg	$n = 279$ Youth team and individual sports athletes from a public sport school 184 males 95 females Mean age 14.6 years (SD 1.7)	*ETL* Number of competitions, intense training days, and light training days *Units* Mean number of intense training days, light training days, and competitions per 100 days during the study period	Cox proportional hazards regression testing adjusted for age and previous injury demonstrated that a higher number of competitions per 100 days was positively associated with injury in team sports athletes (hazard ratio 1.072, 95% CI 1.033–1.113, $p < 0.01$), and negatively associated with injury in individual sports (hazard ratio 0.940 95% CI 0.893–0.989, $p = 0.017$). No association was found between injury and average number of intense or light days per 100 days

Table 1 (continued)

Study	Participants	Training load	Results
Vilamitjana et al., 2013 [75] Prospective cohort study Argentina	$n = 55$ Professional soccer players All males Mean age 23.9 years (SD 3.5)	*ETL* Number of training sessions and matches *Units* Total number of training sessions and matches per season and average number of matches per week	Using independent *t* tests, it was demonstrated that one group of players played significantly more matches per season ($p < 0.01$) and per week ($p < 0.01$), and that this same group had a significantly higher injury IR overall ($p < 0.01$) and in training ($p < 0.01$), but not in matches ($p = 0.1$)
Colby et al., 2014 [52] Prospective cohort study Australia	$n = 46$ Professional and semi-professional Australian rules Football players All males Mean age 25.1 years (SD 3.4)	*ETL* GPS was used to measure distance, speed, and position, and triaxial accelerometers were used to measure acceleration in training sessions and competitions *Units* Several types of ETLs, all reported in AU (1) Velocity load: "a measurement of running power and momentum. The more continuous and higher the velocity equates to a higher velocity load" (2) RVC load: "a calculated function (algorithm) of accelerations, decelerations, and changes of direction, which are summed together to produce an overall 'acceleration load' value." (3) Force load: "a cumulative measurement that sums the forces produced from both foot strikes and collisions. Higher speed running will correspond with higher force load values, reflecting a measure of the 'number' and 'intensity' of foot strikes (i.e. total g-force from foot strikes). In addition, physical contact through collisions and jumping forces will also contribute to force load values"	Multivariate regression analyses were used to compare cumulative and absolute changes in workloads between injured and uninjured players for all ETL-related variables, with the values for each variable split into tertiles. In pre-season, a 3-weekly cumulative velocity load of between 6737 and 8046 AU had a lower injury risk when compared with < 6737 AU (OR 0.239, 95% CI 0.06–0.92, $p = 0.037$). Also in pre-season, a previous to current absolute change in weekly change in RVC load between 0.1 and 9.4 AU recorded a lower injury risk when compared with < 0.1 units (OR 0.040, 95% CI 0.004–0.393, $p = 0.006$). In-season, a cumulative 3-weekly force load > 5397 AU had a greater injury risk compared with < 4651 AU (OR 2.53, 95% CI 1.09–5.87, $p = 0.031$), and a 4-weekly RVC load > 102 AU had a higher injury risk when compared with < 84 AU (OR 2.244, 95% CI 1.06–4.77, $p = 0.035$). No other analyses yielded significant results

Table 1 (continued)

Study	Participants	Training load	Results
Hulin et al., 2014 [18] Prospective cohort study Australia	$n = 28$ Professional cricket fast bowlers All males Mean age 26 years (SD 5)	*ETL and ITL* *ETL* *Absolute* (1) Total number of balls bowled each week in training and in competition Units: bowls/week *Relative* (2) ACWR *ITL* (3) sRPE and duration in minutes for training sessions and competitions (4) ACWR *Units* (1) Number of bowls (2) Ratio of 1-week:4-week average (3) AU (4) Ratio of 1-week:4-week average	*ETL* In a logistic regression model with injury as the dependent variable, and acute workload, chronic workload, and ACWR as the independent variables (all segmented into ordinal variables), it was found that higher acute ETL was associated with lower injury risk in the same week ($p = 0.0001$), but not with injury risk in the subsequent week ($p = 0.172$), and that higher chronic ETL was associated with lower injury risk in the current ($p = 0.002$) and subsequent weeks ($p = 0.017$). An ACWR > 1.0 was associated with an increased risk of injury in the subsequent week (RR 2.1, 95% CI 1.81–2.44, $p = 0.01$) *ITL* In a logistic regression model with injury as the dependent variable, and acute workload, chronic workload, and ACWR as the independent variables (all segmented into ordinal variables), no association was found between acute ($p = 0.176$) or chronic ($p = 0.210$) ITL and injury in the current week. No association was also found between acute ($p = 0.109$) or chronic ($p = 0.381$) ITL and injury in the subsequent week. An ACWR > 1.0 was associated with an increased risk of injury in the subsequent week (RR 2.2, 95% CI 1.91–2.53, $p = 0.009$)
Murray et al., 2014 [59] Prospective cohort study Australia	$n = 43$ Professional rugby players All males Mean age: 24 years (SD 1)	*ETL* Match (session) frequency Recovery time measured as days between matches measured as: Short (5–6 days) Medium (7–8 days) Long (9–10 days)	A Chi-square analysis found no difference between short, medium, and long numbers of days between matches ($p > 0.05$)
Wyss et al., 2014 [72] Prospective cohort study Switzerland	$n = 1676$ Swiss army recruits in basic training All males Mean age 20.7 years (SD 1.2)	*ETL and ITL* *ITL* PAEE per day *ETL* Change in distance covered on foot *ETL* Time spent lifting heavy materials *Units* ITL: MJ/day ETL: change in km/week ETL: minutes of lifting	Multivariate regression analyses demonstrated that increasing PAEE ($p < 0.01$) and time lifting heavy materials ($p < 0.01$) were associated with higher injury IR

Table 1 (continued)

Study	Participants	Training load	Results
Malisoux et al., 2015 [40] Prospective cohort study Location not stated	$n = 517$ Recreational runners 336 males 181 females Mean age: 42.2 years (SD 9.9)	*ETL* Number of sessions Distance covered Time running *Units* Number of sessions per week Distance in km per week Time in hours per week	Cox proportional hazards regression analyses found significantly increased risk of injury for those who ran < 2 vs. ≥ 2 h per week (hazard ratio 3.29, 95% CI 2.27–4.79) and those with an average weekly frequency < 2 vs. ≥ 2 sessions (hazard ratio 2.41, 95% CI 1.71–3.42). The same analyses were also run stratified on BMI, previous injury history, and recent running history. Additional analyses were also run stratified on BMI, previous injury history, and recent running history
Newlands et al., 2015 [89] Prospective cohort study New Zealand	$n = 76$ National rowing team members 46 males 30 females *Mean age* Males 22 years (SD 4) Females 21 years (SD 4)	*ETL* Training duration Distance rowed *Units* Time rowed in min/month Distance rowed in km/month	PPMCC demonstrated a positive correlation between new episodes of LBP and total training hours per month ($r = 0.83$, $p < 0.01$) and average number of km rowed per month ($r = 0.71$, $p = 0.01$)
Orchard et al., 2015 [54] Prospective cohort study Australia	$n = 235$ Professional cricket fast bowlers All males No age information provided	*ETL* Number of overs (a group of six deliveries) *Units* (1) Number of overs delivered during each match (2) Number of overs during the time period leading up to the match	RRs were used to compare future injury occurrence in high vs. low workload bowlers. Fast bowlers who bowled > 50 match overs in a 5-day period had a significant increase in injury over the next month compared with bowlers who bowled ≤ 50 overs or less (RR 1.54, 95% CI 1.04–2.29)

Table 1 (continued)

Study	Participants	Training load	Results
Owen et al., 2015 [71] Prospective cohort study Location not stated	$n = 23$ Professional soccer players All males Mean age 25.6 years (SD 4.6)	*ITL* Intensity measured as HR, measured at 5 s intervals, divided into zones as a percentage of maximum HR (85–90% HRmax: high, ≥ 90% HRmax: very high), duration measured in minutes *Units* Not specified	PPMCCs were used to examine the association between mean time spent at high and very high intensities and injury incidence. A significant relationship was found between overall injury incidence and mean time spent at high ($r = 0.57$, $p = 0.005$) and very high ($r = 0.568$, $p = 0.005$) intensities A stepwise, multiple linear regression analysis was used to generate OR to determine if mean time spent at high or very high HR was associated with frequency of injury in practice and competition. For high intensity, the ORs were 0.60 (95% CI 0.28–1.28, $p = 0.20$) for practice, and 1.87 (95% CI 1.12–3.12, $p = 0.02$) for competition. For very high intensity, the ORs were 0.52 (95% CI 0.24–1.12, $p = 0.10$) and 1.28 (95% CI 0.78–2.10, $p = 0.38$) for competition *T* tests were used to determine if time spent in high and very high intensities differed between those who sustained and did not sustain an injury in the following month. For high intensity, the difference was non-significant ($p = 0.06$); for very high intensity, the difference was significant ($p = 0.04$)
Bayne et al., 2016 [77] Prospective cohort study Australia	$n = 25$ District and/or state cricket squad players All males Mean age 15.8 years (range 14–19)	ETL measured in several ways Average number of overs per week, average number of sessions per week, average number of overs per session, maximum number of overs in a single session, maximum number of overs in a single week, maximum number of overs in any 2-week period, and maximum number of overs in any 4-week period *Units* Number of overs or sessions in a specified time period	Independent *t* tests were used to assess for differences in all training load variables between injured and non-injured participants. No significant differences were found for any training load variable (no *p* values reported)

Table 1 (continued)

Study	Participants	Training load	Results
Carey et al., 2017 [66] Prospective cohort study Australia	$n = 53$ Professional Australian rules Football players All males Mean age 22.9 years (SD 4.0)	*ETL and ITL* *ETL* *Absolute* (1) Total distance covered in training and competition (2) Distance covered above 24 km/h (3) Distance covered between 18 and 24 km/h *Relative* (4) ACWR *Units* (1) Distance in m (2) Distance in m (3) Distance in m (4) 1-week:4-week training load ratio *ITL* *Absolute* (1) sRPE and duration in minutes for training sessions and competitions *Relative* (2) ACWR *Units* (1) AU (2) 1-week:4-week training load ratio	Quadratic regression equations modeling workload assessed as ETL and ITL found significant associations between greater ACWR and injury risk, with the greatest model R2 found for ETL measured as moderate running distance

Table 1 (continued)

Study	Participants	Training load	Results
Hulin et al., 2016 [56] Prospective cohort study Australia	$n = 53$ Professional rugby players All males Mean age 23.4 years (SD 3.5)	*ETL* (1) Distance covered in practices and competitions measured using *GPS* (2) ACWR *Units*: (1) Distance in m (2) Ratio of 1-week:4-week average	Absolute ETL: Acute and chronic absolute workloads were transformed into ordinal variables with six levels, ranging from very low to very high. Using this scheme, a very high acute workload (≥28,798 m) was associated with an increased risk of injury relative to the five other ordinal levels (RR 1.9–13.9, ×/÷1.9–2.2; range 86–98%, likely to very likely), but *p* values were not reported. Instead, calculated *p* values were converted to probabilities of the effect being beneficial or harmful, with practical significance being claimed when the probability of benefit or harm was ≥75%. No difference was found in injury risk between any of the six ordinal levels of chronic absolute workload *Relative ETL* ACWRs were calculated using the continuous absolute ETL measured and then converted into six ordinal categories ranging from very low to very high. In the current week, the very high category (≥2.11) had an injury risk 6.9 times that of the very low category ≤0.30 (RR 6.9×/÷1.7; likelihood = 98%, very likely), 3.4 times greater than a low ratio of 0.31–0.66 (RR 3.4×/÷2.0; likelihood 97%, very likely), 2.3 times greater than a moderate ratio of 1.03–1.38 (RR 2.3×/÷3.4; likelihood 91%, likely), and 2.0 times that of a high ratio of 1.75–2.10 (RR 2.0×/÷17.2; likelihood 77%, likely)
Kluitenberg et al., 2016 [39] Prospective cohort study The Netherlands	$n = 1696$ Recreational runners 364 males 1332 females Mean age 43.3 years (SD 10)	*ETL and ITL* *ITL* (1) sRPE on a 6–20 scale and duration in minutes for training sessions and competitions *ETL* (2) Running session frequency *Units* (1) AU (2) Number of sessions per week	In a multivariate extended Cox regression model, higher intensity was associated with injury occurrence (hazard ratio 1.28 95% CI 1.18–1.40) and running more than 60 min in the previous 7 days was protective (hazard ratio 0.41, 95% CI 0.20–0.86). No association was found between running frequency and injury [greatest hazard ratio (found between two and three sessions) 1.42, 95% CI 0.97–2.08]

Table 1 (continued)

Study	Participants	Training load	Results
Krutsch et al., 2016 [90] Prospective cohort study Germany	$n = 408$ Professional and elite amateur soccer players All males Mean age: 26 years (SD 4)	*ETL* Training duration *Units* Total hours of training	A *t* test demonstrated that the professional teams had a significantly higher mean training time. An exact test based on the Poisson distribution found no significant differences in overall injury rate between the professional and amateur teams (IRR 0.88, 95% CI 0.74–1.04). However, a higher rate of cruciate ligament injuries was found in the professional (higher load) group (IRR 3.38, 95% CI 1.03–13.50, $p = 0.04$)
Murray et al., 2017 [60] Prospective cohort study Australia	$n = 46$ Professional Australian rules Football players All males Mean age 23.1 years (SD 3.7)	*ETL* Pre-season session participation frequency *Units* Percentage of pre-season sessions completed	Chi-square tests used to compare differences in injury IR between players with low (< 50% pre-season session participation), moderate (participation in 51–85% of sessions), and high (participation in > 85% of sessions) training loads during the pre-season demonstrated no significant differences between the groups ($p = 0.17$)
Orr and Cheng 2016 [91] Prospective cohort study Australia	$n = 368$ Elite amateur junior rugby players All males Mean age 15.8 years (SD 1.0)	*ETL* Duration of weekly physical activity *Units* Hours spent in team training, additional sporting endeavors, exercise, or work/manual labor per week	A Spearman's correlation found no association between weekly physical activity and injury incidence expressed as either (1) absolute number of match injuries ($p = -0.03$; $p = 0.83$), or (2) match injuries per hour ($p = -0.07$; $p = 0.60$)
Thornton et al., 2016 [92] Prospective cohort study Australia	$n = 25$ Professional rugby players All males No age information provided	ETL and ITL measured in multiple ways *ETL* (1) Total distance covered (2) Total high-velocity distance covered (3) Total high-metabolic power distance covered *ITL* (1) sRPE *Units* *ETL* (1) Distance covered in m (GPS) (2) Distance covered at > 5 m/s (GPS) (3) Distance covered at > 20 W/kg (GPS) *ITL* (1) AU	Multivariate binary logistic regression with GEEs found significant associations between several measures of both ETL and ITL for various player positions
Van der Worp et al., 2016 [93] Prospective cohort study The Netherlands	$n = 435$ Recreational runners All females Mean age 38.7 years (SD 11.5)	*ETL* Weekly running distance *Units* Km run per week	A multivariate Cox regression model found that those runners with a weekly distance > 30 km were significantly more likely to incur an injury, adjusting for several other potential risk factors (hazard ratio 3.28, 95% CI 1.23–8.75, $p = 0.02$)

Table 1 (continued)

Study	Participants	Training load	Results
Ball et al., 2018 [35] Prospective cohort study Australia	$n = 51$ Amateur (university) rugby players All males Mean age 19.2 years (SD 0.7)	*ITL* sRPE and duration in minutes for training sessions and competitions *Units* AU	In forward stepwise multiple regression analyses, a lower training load was associated with a higher injury incidence in 'back' position players ($p < 0.01$), but not in 'forward' position players ($p > 0.05$), adjusted for one repetition maximums for bench press and deadlift
Bowen et al., 2017 [67] Prospective cohort study UK	$n = 32$ Elite youth soccer players Sex not stated Mean age 17.3 years (SD 0.9)	ETL measured in five ways GPS was used to measure distance, speed, and position, and triaxial accelerometers were used to measure acceleration in training sessions *Absolute* (1) Total distance (2) Distance covered above speed of 20 km/h (3) Total forces acting on each player over the entire season (based on accelerometer data) (4) A change in GPS speed data for at least 0.5 s, with maximum acceleration in the period of at least 0.5 m/s/s *Relative* (5) ACWR *Units* (1) Distance in m (2) Distance in m (3) AU (4) Number of accelerations (5) 1-week:previous 4-week ratios of variables (1–4)	Binary logistic regression analyses were used to compare all workload variables between injured and uninjured participants. A weekly high-speed distance of moderate–high (856–1448 m) was associated with a greater overall injury risk (RR 1.73, 95% CI 1.06–2.84, $p = 0.029$). A weekly total load of 474–648 AU was associated with a greater overall risk of injury (RR 1.65, 95% CI 1.04–2.62, $p = 0.032$). A low weekly total load (0–130 AU) was associated with significantly reduced overall injury risk (RR 0.27, 95% CI 0.12–0.60, $p = 0.002$) A low ACWR (0–0.36) for 4-weekly total high-speed distance covered significantly reduced the overall injury risk (RR 0.47, 95% CI 0.25–0.9, $p = 0.022$). No significant associations were found between the ACWR for total load and overall injury risk
Colby et al., 2017 [36] Prospective cohort study Australia	$n = 70$ Professional Australian rules Football players All males Mean age: 22.9 years (SD 2.4)	ETL measured two ways (1) Total distance covered in practices and competitions measured using GPS (2) Sprint distance: the distance covered at > 75% of a player's maximum speed *Units* (1) Distance in m (2) Distance in m	Multivariate binary logistic regression with GEE adjusted for age, previous injury, height and weight found that very low loads (in terms of total distance) in the late pre-season was associated with greater risk of injury during the in-season period compared with moderate loads (OR 5.6, 95% CI 1.4–22.6, $p = 0.02$). No significant association was found between sprint distance and injury risk

Table 1 (continued)

Study	Participants	Training load	Results
Colby et al., 2017 [37] Prospective cohort study Australia	$n = 70$ Professional Australian rules Football players All males Mean age 22.9 years (SD 3.4)	ETL and ITL measured in multiple ways: *ETL* *Absolute* (1) Total distance covered in training and competition (2) Sprint distance: the distance covered at > 75% of a player's maximum speed *Relative* (3) ACWR (ACWR) *Units* (1) Distance in m (2) Distance in m (3) Ratio of 1-week:4-week average *ITL* *Absolute* (1) sRPE and duration in minutes for training sessions and competitions *Relative* (2) ACWR *Units* (1) AU (2) Ratio of 1-week:4-week average	Multivariate mixed-model GEE modeling demonstrated that a low chronic ETL (total distance) coupled with a high ACWR was associated with an increased risk of injury (adjusted IRR 2.60, 95% CI 1.07–6.34) compared with an average chronic distance with a moderate ACWR. A similar model showed that a low chronic ITL with a low ACWR was associated with increased risk (adjusted IRR 2.52, 95% CI 1.01–6.29) compared with an above-average chronic load and moderate ACWR
Harrison and Johnston, 2017 [61] Prospective cohort study Australia	$n = 60$ Semi-professional Australian rules Football players All males Mean age 21.3 years (SD 2.9)	*ITL* *Absolute* sRPE and duration in minutes for training sessions *Units* AU *Relative* Week-to-week changes in total ITL *Units* Week-to-week difference, expressed as a percentage	Binary logistic regression demonstrated that participants who had 2-week loads > 4000 AU were significantly more likely to be injured in the subsequent week compared with those with moderate loads (2000–4000 AU) [OR 2.80, 95% CI 1.05–7.45, $p = 0.04$). In addition, players with an increase in load of 15–49% of the previous week had a significantly higher risk of injury the following week (OR 3.76, 95% CI 1.30–10.86, $p = 0.01$)

Table 1 (continued)

Study	Participants	Training load	Results
Malone et al., 2017 [63] Prospective cohort study Ireland	$n = 37$ Elite amateur Gaelic Football players All males Mean age 24 years (SD 6)	*ITL* (1) sRPE and duration in minutes for training sessions and competitions (2) ACWR *Units* (1) AU (2) Ratio of 1-week:4-week average	Binary logistic regression demonstrated that participants who exerted in-season weekly loads of > 1500 to 2700 AU were at significantly higher risk of injury compared with the reference group of < 1200 AU (OR 2.44, 95% CI 1.28–4.66, $p = 0.006$), and that players who had in-season 2- and 3-weekly loads of > 3950 and > 4750 AU were at significantly higher risk of injury compared with the reference group of < 1950 AU (OR 4.74, 95% CI 1.14–8.76, $p = 0.033$) and < 2750 AU (OR 6.11, 95% CI 4.26–8.14, $p = 0.023$). Participants who performed previous to current week changes > 1000 AU in the pre-season and late in-season phase were at significantly higher risk compared with the reference group of < 120 AU (OR 2.58, 95% CI 1.09–4.52, $p = 0.001$) (OR 7.42, 95% CI 6.41–8.12, $p = 0.002$). Participants who exerted in-season ACWRs of > 1.35 to 1.50 (OR 0.88, 95% CI 0.28–4.66, $p = 0.006$) were at significantly lower risk of injury compared with the reference group of < 1.00. Participants who exerted in-season ACWRs of > 2.00 were at increased injury risk (OR 3.33, 95% CI 1.69–6.65, $p = 0.001$) compared with the reference group of < 1.00
Malone et al., 2017 [62] Prospective cohort study Europe	$n = 48$ Professional soccer players All males Mean age 25.3 years (SD 3.1)	*ITL* (1) sRPE and duration in minutes for training sessions and competitions (2) ACWR using sRPE (3) Week-to-week changes in absolute workload *Units* (1) AU (2) Ratio of 1-week:4-week average (3) AU	Binary logistic regression analyses comparing the absolute workloads between injured and uninjured players demonstrated that players who exerted pre-season weekly workloads of ≥ 1500 to ≤ 2120 AU were at significantly higher risk of injury compared with the reference group of ≤ 1500 AU (OR 1.95, 95% CI 0.98–3.95; $p = 0.006$), and players who had completed a pre-season 2- and 3-weekly workload of ≥ 5980 and ≥ 9154 AU were at significantly higher risk of injury compared with the reference group of < 3250 AU (OR 4.74, 95% CI 2.74–5.66, $p = 0.033$) and < 7260 AU (OR 5.11, 95% CI 4.26–5.14, $p = 0.023$) Binary logistic regression examining relative workload found players who exerted in-season ACWRs of > 1.00 to < 1.25 were at significantly lower risk of injury compared with those ≤ 0.85 (OR 0.28, 95% CI 0.08–1.26, $p = 0.001$)

Table 1 (continued)

Study	Participants	Training load	Results
Malone et al., 2018 [49] Prospective cohort study Portugal	$n = 37$ Elite soccer players All males Mean age: 25 years (SD 3)	*ETL and ITL* *ETL (GPS)* *Absolute* (1) Distance covered above 14.4 km/h (HSR) (2) Distance covered above 19 km/h (SR) *Relative* (3) Week-to-week changes in absolute load (4) ACWR *Units* (1) Distance in m (2) Distance in m (3) Difference in m between 2 successive weeks (4) Ratio of 3- to 21-day workload total in m (unitless) *ITL* *Absolute* (1) sRPE and duration in minutes for training sessions and competitions *Relative* (2) week-to-week changes in absolute load (3) ACWR *Units* (1) AU (2) Difference in workload between two successive weeks in AU (3) Ratio of 3 day to 21 day workload (unitless)	Binary logistic regression demonstrated that players who completed moderate HSR (701–750 m: OR 0.12, 90% CI 0.08–0.94, $p = 0.025$) and SR distances (201–350 m: OR 0.54, 90% CI 0.41–0.85, $p = 0.005$) were at reduced injury risk compared with low HSR and SR groupings (HSR ≤ 674 m; SR ≤ 165 m) and the high (HSR between 750 and 1025 m; SR 350–525 m) reference groups; Injury risk was greater for players who experienced large weekly changes in HSR (351–455 m; OR 3.02, 90% CI 2.03–5.18, $p = 0.011$) and SR distances (75–105 m; OR 6.12, 90% CI 4.66–8.29; $p = 0.001$) compared with the reference HSR (≤ 100 m) and SR (≤ 50 m) group; players who had an HSR 3:21-day ACWR 1.25, and a 3:21-day SR distance ACWR of > 1.35 were at increased risk of subsequent injury; players who exerted higher 21-day chronic training loads (≥ 2584 AU) were at reduced risk of injury when they covered weekly HSR distances of 701–750 m compared with the reference group of < 674 m (OR 0.65, 90% CI 0.25–0.89, $p = 0.024$). Conversely, players who exerted low chronic training loads (≤ 2584 AU) and covered the same distance of 701–750 m were at greater risk of injury compared with the reference group of < 674 m (OR 3.12, 90% CI 2.99–4.54, $p = 0.036$)
Malone et al., 2017 [48] Prospective cohort study Ireland	$n = 37$ Elite amateur Gaelic football players All males Mean age 24 years (SD 3)	*ETL and ITL* *ETL* (1) Maximum velocity exposure *Units* (1) Meters covered at individual's maximum velocity *ITL* (1) sRPE and duration in minutes for training sessions and competitions (2) ACWR *Units* (1) AU (2) Ratio of 1-week:4-week average	Binary logistic regression demonstrated that players who reached at least 95% of their maximum velocity during training or competition were significantly less likely to incur an injury the subsequent week (OR 0.12, 95% CI 0.01–0.92, $p = 0.001$); players with a lower chronic training load (≤ 4750 AU) were at increased injury risk (OR 1.44, 95% CI 1.28–2.22, $p = 0.107$) when exposed to 10–15 maximum velocity events compared with those with ≥ 4750 AU, while those with higher chronic loads were at reduced risk (OR 0.23, 95% CI 0.10–1.33, $p = 0.055$)

Table 1 (continued)

Study	Participants	Training load	Results
Moller et al., 2017 [65] Prospective cohort study Denmark	$n = 679$ Elite youth handball players 380 males 299 females Age range 14–18 years	*ETL* Change in weekly hours of handball training and competition *Units* *Relative* Week-to-week change in hours of training/competition	Cox proportional hazards regression models with frailty found no difference in injured vs. uninjured players with week-to-week changes of < 20, 20–60, and > 60%, controlling for scapular control, isometric rotational and abduction strength, and shoulder range of motion (p value ranges 0.09–0.68)
Murray et al., 2017 [55] Prospective cohort study Australia	$n = 59$ Professional Australian rules Football players All males Mean age 24 years (SD 3)	*ETL* *Absolute* (1) Total distance covered (2) Total low-speed distance covered (3) Total moderate-speed distance covered (4) Total high-speed distance covered (5) Total very-high-speed distance covered 6) Player load defined as instantaneous rate of change in *X*/*Y*/*Z* axes (accelerometer) *Relative* (7) Rolling averages *ACWR* (8) EWMA ACWR *Units* (1) Distance covered in m (GPS) (2) Distance covered at < 6 km/h (GPS) (3) Distance covered at 6–18 km/h (GPS) (4) Distance covered at 18–24 km/h (GPS) (5) Distance covered at > 24 km/h (GPS) (6) AU (7) Ratio of 1-week total to 4-week average total workloads (8) Ratio of 1-week total to 4-week total using a formula that accounts for 'decay' in fitness and fatigue by diminishing the weight of older load measures of both the acute and chronic loads	Binary logistic regression demonstrated that both the rolling average ACWR and the EWMA ACWR were both significantly associated with injury risk, with ACWR values > 2.0 associated with significantly greater risk of injury compared with those with values of 1.0–1.49 across a series of loading variables in both models

Table 1 (continued)

Study	Participants	Training load	Results
Stares et al., 2017 [64] Prospective cohort study Australia	$n = 70$ Professional Australian rules Football players All males Mean age: 22.9 years (SD 3.4)	*ETL and ITL* *ETL* *Absolute* (1) Total distance covered in training and competition (2) Sprint distance: the distance covered at > 75% of a player's maximum speed *Relative* (3) ACWR *Units* (1) Distance in m (2) Distance in m (3) Ratio of 1-week:4-week average *ITL* *Absolute* sRPE and duration in minutes for training sessions and competitions *Units* AU *Relative* ACWR *Units* Ratio of 1-week:4-week average	A mixed-model GEE adjusting for age, experience, and with a chronic load–ACWR interaction term demonstrated that the ACWR from both ETL and ITL was significantly associated with injury risk ($p < 0.05$)
Weiss et al., 2017 [46] Prospective cohort study New Zealand	$n = 13$ Professional basketball players All males Mean age 24.4 years (SD 4.7)	*ITL* *Relative* ACWR formed through sRPE and duration in minutes for training sessions *Units* 1-week:4-week ratio of training load	Injury IP ratios comparing injury incidence in the following week between players with various ACWRs, using the 90% CIs used to determine significance, found that those with a ratio between 1.0 and 1.49 were significantly less likely to be injured than those with higher and lower ratios
Williams et al., 2017 [47] Prospective cohort study UK	$n = 1253$ Professional rugby players All males Mean age 26 years (SD 4)	*ETL* Match exposures *Units* Number of matches in which a player participated for at least 20 min during the preceding 1- and 12-month period	Cox proportional hazards regression with frailty found that those players exposed to > 1-month match exposure were significantly more likely to experience injury (hazard ratio 1.14 per 2 SDs increase, 90% CI 1.08–1.20). The authors also reported that players with fewer than 15 and more than 35 matches in the preceding 12 months were significantly more likely to be injured than those with moderate exposure, adjusted for age, height, mass, position, and previous injuries, although no formal statistics were reported

ACWR acute chronic workload ratio, *ANOVA* analysis of variance, *AU* arbitrary units, *BMI* body mass index, *CI* confidence interval, *ETL* external training load, *EWMA* exponentially weighted moving average, *GEE* generalized estimating equation, *GPS* global positioning system, *HR* heart rate, *HRmax* maximum heart rate, *HSR* high-speed running, *IP* injury proportion, *IR* injury rate, *IRR* injury rate ratio, *ITL* internal training load, *LBP* low back pain, *OR* odds ratio, *MJ* megajoules, *NCAA* National (US) Collegiate Athletic Association, *PAEE* physical activity energy expenditure, *PPMCC* Pearson product moment correlation coefficient, *R2* coefficient of determination, *RR* risk ratio, *RVC* relative velocity change, *SD* standard deviation, *SR* sprint running, *sRPE* session rating of perceived exertion, *VO2max* volume of maximal oxygen consumption

relationship [35–41], four reported finding both inverse and direct relationships [18, 42–44], three reported a U-shaped relationship [45–47], two reported finding a U-shaped and inverse relationship [48, 49], and one reported finding a direct and an inverted U-shaped relationship [50]. 12 studies identified a delayed window of risk for injury after an increase in load [18, 40, 51–60], 12 identified a protective effect of moderate absolute loads [35–37, 41, 45, 47–50, 52, 61, 62], and 11 reported protective effects from moderate relative changes in loading [18, 37, 46, 48, 49, 55, 56, 61–64]. All articles except two [55, 65] utilized an absolute measure of training load. Twelve articles used the ACWR [18, 37, 46, 48, 49, 55, 56, 62–64, 66, 67] and six used week-to-week changes in training load [49, 57, 61–63, 65].

A total of 43 articles measured at least one form of ETL. Of these, 15 utilized session duration, 16 utilized session frequency, 13 utilized distance, and six utilized number of repetitions to quantify load. Thirty-three of the 40 studies that utilized ETL had at least partially statistically significant results, with 21 reporting all statistically significant results. Furthermore, 19 studies that utilized ETL had at least partially null results, with ten reporting all null results.

Of the 25 studies that utilized some form of ITL, 22 used a subjective measure and three utilized an objective measure. Twenty-four of the 25 had at least partially statistically significant results. Nineteen had all statistically significant results, five had mixed results, and one [68] reported all null results. All 22 of the articles that examined subjective ITL utilized sRPE multiplied by session duration, with 21 of these using the 10-point modified Borg scale adapted by Foster et al. [17] to quantify session intensity, and one [69] utilizing a modified version of this scale. All but one study that utilized sRPE reported statistically significant results, with 18 reporting all statistically significant results and only one [68] reporting all null findings. Two of the three articles that used objective measures of ITL used HR [70, 71], and one used estimated energy expenditure [72]. Two of the articles that measured objective ITL reported all statistically significant results [70, 72] and one reported mixed results [71].

Of the 15 articles that examined a relative measure of training load, three used week-to-week changes in load, nine used the ACWR, and three used both. The ACWR is most often defined as the ratio of a week's training load to the mean of the previous four weeks [18]. For example, if a week's training load was 500 AU and the mean training load of the previous 4 weeks was 250 AU, the ACWR for that week would be 2.0. All but one [65] of the articles that examined relative workload had at least some statistically significant results, including all statistically significant results reported in ten of the 12 articles that used the ACWR. The one article that reported all null results used week-to-week changes in session duration as the measure of load [65].

3.4 Assessment of Article Quality, Level of Evidence, and Conflicts of Interest

The median overall NOS score was 6 (range 4–7), the median participant selection score was 3 (range 2–4), the median comparability score was 1 (range 0–2), and the median outcome score was 2 (range 0–2). An article's score on the NOS can be translated to 'good', 'fair', and 'poor' levels of article quality per the US Agency for Healthcare Research and Quality (AHRQ) guidelines (electronic supplementary material Appendix S2) [29, 73]. Included articles were rated as poor ($n = 15$), fair ($n = 6$), and good ($n = 36$). The 36 articles that were assessed as 'good' were considered level 2b evidence on the Oxford Centre for Evidence-based Medicine Model, while the 21 articles assessed as 'fair' or 'poor' were considered level 4 evidence [33]. Scores for each article are presented in Table 2. Twenty-four of the included studies reported receiving funding to complete the research in full or part. The authors of 28 studies stated they had no potential conflicts of interest, two reported a potential conflict of interest [53, 54], and the authors of 27 studies did not include a statement as to whether or not there was a potential conflict of interest in completing their study.

4 Discussion

The primary purpose of this systematic review was to examine the evidence for a relationship between training load and musculoskeletal injury risk, taking recently published studies into account, broadening the definition of training load, and expanding the definition of athletes to include tactical athletes.

In their reviews, both Jones et al. [19] and Drew and Finch [20] concluded there was 'emerging moderate evidence' for a relationship between training load and injury risk. Since the publication of the review by Drew and Finch [20], which featured the most recent literature search (October 2015), 24 additional articles examining the injury-load relationship in athletes have been published. Twenty of these reported significant findings for a relationship between training load and musculoskeletal injury. Ten utilized a measure of ETL, seven utilized a measure of ITL, and seven used both. Eleven of these articles utilized a measure of absolute load, two examined relative load, and 11 used both. Together, these studies have contributed meaningfully to the literature by strengthening the evidence for a relationship between training load and injury risk, and have led us to conclude that the evidence for the existence of this relationship has shifted from emerging to established. However, it is important to note that this evidence still rates as a 'moderate' level using the Van Tulder et al. method, owing to a lack of RCTs. The recently published studies also provide some unique insights

Table 2 Quality of included studies as assessed on the Newcastle–Ottawa Scale (NOS)

Study	NOS Score				Level of evidence[a]
	Selection	Comparability	Outcome	Total score	
Ekstrand et al. 1983 [50]	2	1	1	4	4
Korkia et al. 1993 [76]	2	1	1	4	4
Hamalainen et al. 1994 [79]	3	1	2	6	2b
Anderson et al. 2003 [82]	3	1	2	6	4
Dennis et al. 2003 [45]	3	2	2	7	2b
Taunton et al. 2003 [41]	2	2	2	6	4
Dennis et al. 2004 [51]	3	1	2	6	2b
Gabbett 2004 [88]	3	2	2	7	2b
Gabbett 2004 [83]	3	2	2	7	2b
Gabbett and Domrow 2005 [38]	2	2	1	5	4
Gabbett and Domrow 2007 [84]	2	1	1	4	4
Brooks et al. 2008 [85]	3	1	2	6	2b
Gianoudis et al. 2008 [86]	2	0	2	4	4
Orchard et al. 2009 [53]	3	1	2	6	2b
Brink et al. 2010 [58]	3	1	2	6	2b
DuPont et al. 2010 [74]	3	1	2	6	2b
Killen et al. 2010 [68]	3	1	2	6	2b
Wilson et al. 2010 [87]	2	1	1	4	4
Gabbett and Jenkins 2011 [88]	3	2	2	7	2b
Gabbett and Ullah 2012 [43]	4	2	2	8	2b
Mallo and Dellal 2012 [70]	3	2	2	7	2b
Colby et al. 2014 [52]	3	2	2	7	2b
Jacobsson et al. 2013 [69]	3	2	2	7	2b
Rogalski et al. 2013 [57]	3	2	2	7	2b
Theisen et al. 2013 [44]	2	2	2	6	4
Vilamitjana et al. 2013 [75]	2	1	2	5	4
Hulin et al. 2014 [18]	3	1	2	6	2b
Murray et al. 2014 [59]	4	1	1	6	4
Wyss et al. 2014 [72]	3	2	1	6	4
Dellal et al. 2015 [42]	3	1	2	6	2b
Malisoux et al. 2015 [40]	2	1	1	4	4
Newlands et al. 2015 [89]	3	1	1	5	4
Orchard et al. 2015 [54]	3	1	2	6	2b
Owen et al. 2015 [71]	3	2	2	7	2b
Bayne et al. 2016 [77]	4	1	1	6	4
Carey et al. 2016 [66]	3	2	2	7	2b
Hulin et al. 2016 [56]	3	1	2	6	2b
Kluitenberg 2016 [39]	3	1	0	4	4
Krutsch et al. 2016 [90]	3	1	2	6	4
Murray et al. 2016 [60]	4	1	2	7	2b
Orr and Cheng 2016 [91]	2	1	2	5	4
Thornton et al. 2016 [92]	2	2	2	6	4
Van der Worp et al. 2016 [93]	2	2	2	6	4
Ball et al. 2017 [35]	3	2	2	7	2b
Bowen et al. 2017 [67]	3	2	2	7	2b
Colby et al. 2017 [36]	3	2	2	7	2b
Colby et al. 2017 [37]	3	2	2	7	2b
Harrison et al. 2017 [61]	3	1	2	6	2b
Malone et al. 2017 [63]	3	1	2	6	2b

Table 2 (continued)

Study	NOS Score				Level of evidence[a]
	Selection	Comparability	Outcome	Total score	
Malone et al. 2017 [62]	3	1	2	6	2b
Malone et al. 2017 [49]	3	2	2	6	2b
Malone et al. 2017 [48]	3	2	2	7	2b
Moller et al. 2017 [65]	3	2	2	7	2b
Murray et al. 2017 [55]	3	1	2	6	2b
Stares et al. 2017 [64]	3	2	2	7	2b
Weiss et al. 2017 [46]	3	1	1	5	4
Williams et al. 2017 [47]	3	2	2	7	2b
Median (range)	3 (2–4)	1 (0–2)	2 (0–2)	6 (4–7)	–

[a] As determined by the Oxford Centre for Evidence-based Medicine

2b: Low quality randomized controlled trial or cohort study with good reference standards

4: Case series or poor quality prognostic cohort study

into the load–injury relationship, including that ratios other than the original 1:4 weekly version of the ACWR are significantly associated with injury risk [49, 64, 66], and demonstrating that an ACWR calculated daily instead of weekly is significantly associated with injury risk [55, 66].

Drew and Finch also made several other observations and conclusions in their review, including that there was no level 1 evidence in the form of RCTs, that sRPE was the most commonly used measure in the literature, that both absolute and relative loads appear to be associated with injury risk, that increases in injury risk may be delayed after increases in loading, and that there appears to be a protective effect of moderate loading [20]. Our findings largely agree with those of Drew and Finch. Our search strategy also yielded no level 1 evidence and found the most frequently used measure of training load was sRPE. We also found good evidence of a protective effect of moderate absolute loads [35–37, 41, 45, 47–50, 52, 61, 62] and moderate relative changes in load [18, 37, 46, 48, 49, 55, 56, 61–64], as well as a delayed window of injury vulnerability after an increase in load [18, 40, 51–60].

Expanding upon and updating the previous reviews [19, 20], our definition of training load included session frequency (e.g. matches per week), along with measures such as distances, durations, and repetitions. We elected to define training load broadly to ensure that the most complete picture possible of the relationship between injury and training load, as defined in the recent IOC consensus statement on training load monitoring [1], could be crafted in this review. We found conflicting evidence for a relationship between session frequency and injury risk. Of the 16 studies examining this relationship, two studies reported a significantly increased risk of injury with greater session frequency (or fewer rest days) [74, 75], five studies reported an increased risk with underloading in session frequency [38, 40, 45, 47, 50], six reported no association [39, 59, 60, 70, 76, 77], and three reported mixed results [41, 42, 44]. Taken together, these findings suggest the strength of evidence for session frequency using the Van Tulder et al. method is 'conflicting' [34]. Interestingly, this was the conclusion of Jones et al. in their review, which examined only six of the 16 articles included in this review that utilized session frequency. Jones and Drew also found direct, indirect, mixed, and no relationships in studies using session frequency. Congruence between these two reviews suggests that despite ten additional studies examining this relationship, the directionality and significance of findings have not meaningfully changed, and thus the association between session frequency and future injury risk may be too weak for it to serve as a useful predictor of injury in isolation.

A significant challenge remaining in this area of research is determining what factors influence the direction of the relationship between load and injury risk. While a majority of studies reviewed reported a direct relationship between load and injury risk, others reported finding inverse or U-shaped relationships. One important influence on this relationship is the chronicity of load measures used. The majority of studies reporting a direct relationship utilized an acute absolute measure of load, while the majority of those reporting an inverse relationship utilized a measure of chronic load (e.g. load of the previous 4 weeks) [18, 37, 43, 44, 48, 49, 52]. These findings align with Gabbett's 'training load–injury paradox', which states that athletes accustomed to high chronic loads have fewer injuries than those accustomed to lower loads, and supports Gabbett's assertion that higher chronic loads can act as a protective 'vaccine' against future injury. Studies reporting a U-shaped relationship between load and injury utilized measures of ITL [46] and ETL [45, 47–49], as well as absolute [45, 47–50] and relative [46] load. The results of these studies suggest

that low loads may not be enough to elicit protective physiologic adaptations in athletes, while high loads may result in overloading of tissue or decreased injury resistance by some other mechanism. Similarly, studies examining relative loads demonstrated that small to moderate changes relative to previous loads were associated with a reduced risk of injury compared with larger or smaller changes [37, 46, 48, 49, 55, 63, 64]. This corresponds to Gabbett's concept of a loading 'sweet spot' in which acute loading is similar to chronic loading [78]. Together, this body of evidence suggests that to minimize injury risk, athletes should receive gradually increasing doses of loading during the off-season and pre-season, with the goal of obtaining a protective 'vaccine' that results from having a high chronic load to prevent injuries during the competition period [78], although this must also be balanced against load scheduling for performance optimization.

In addition to traditional, or competitive, athletes, this review evaluated the strength of evidence of a relationship between training load and injury risk in tactical athletes. Our review found only two studies that prospectively examined the relationship between training load and injury risk in tactical athletes. Hamalainen et al. reported greater incidence of acute in-flight neck pain in fighter pilots with longer mean flight times [79]. In a study of Swiss military recruits, Wyss et al. found that increasing physical activity energy expenditure and time lifting heavy materials were both directly associated with injury incidence [72]. Although few in number, these studies suggest that the training load–injury risk relationship seen in competitive athletes exists in tactical athletes, indicating that the methods used to monitor training load in competitive athletes may also be useful in tactical athletes. However, this relationship may differ slightly due to several unique non-activity-related characteristics of tactical athlete populations, such as greater ranges of age and physical fitness level, greater variance in experience with tasks performed and training history, and significant variation in environment and health conditions (e.g. sleep deprivation, prolonged cold/heat exposure, greater psychological stressors) [21].

Comparing the outcomes of studies that used ITL versus ETL to accomplish our secondary aim, the literature utilizing ITL had a greater proportion of studies with at least some significant findings (23/25 vs. 33/43). This is especially true for studies utilizing sRPE. Twenty-one of 22 studies using sRPE reported significant findings. The only study that used sRPE and had null findings was the study by Killen et al. [68], which had a small sample size ($n = 36$) and an unusually low injury incidence rate (6.9 per 1000 training hours), likely resulting in it being underpowered. This study also utilized a short (one pre-season) observation period. While the proportion of studies using ETL reporting significant results was not as high as that using ITL, a plurality of those with null findings came from studies using session duration to measure load. Of these, fewer than half (7/15) had significant findings. When these are removed, the proportion of studies utilizing ETL with significant findings more closely resembles ITL.

The overall quality of included studies was fairly robust. A majority ($n = 36$) were rated as 'good' on the NOS [29], and these were considered level 2b evidence on the Oxford Centre for Evidence-based Medicine levels of evidence Model, the highest possible level for individual cohort studies [33]. The studies ranked as 'fair' ($n = 6$) and 'poor' ($n = 15$) were considered level 4 evidence on the Oxford Centre for Evidence-based Medicine Model. The most frequently missed criteria were failure to demonstrate that the outcome of interest was not present at the start of the study, controlling for pertinent factors (e.g. sex, previous injury history, years of playing experience), and enumeration of dropouts. Future studies should be sure to address these issues to ensure high methodological quality.

Despite a majority of studies in this review being rated as 'good' quality, the overall strength of evidence for the load–injury relationship is moderate, as assessed by the Van Tulder et al. levels of evidence method [34]. The reason for this is the lack of RCTs on this topic; multiple RCTs are required for the evidence to be assessed as 'strong'. We recommend future research on this topic includes RCTs or cluster RCTs consisting of an active arm of individuals or teams that receive load-attenuating interventions when loads either approach a pre-specified absolute or relative load 'spike', such as an ACWR ≥ 1.50, and a control arm receiving standard load monitoring but no intervention. Future studies should also seek to go beyond examining an association in the load–injury relationship and assess the predictive properties of various measures of ETL and ITL in combination with other risk factors.

This review has several strengths. First, it was conducted using PRISMA guidelines. Second, the search strategy utilized to identify articles of interest was comprehensive and included identifying articles not found in the search but known to the authors or cited in studies found in the search, ensuring a virtually complete capture of relevant studies. Third, the broad definition of training load used, including session frequency, ensured that assessment of all dimensions of the load–injury risk relationship was completed. Fourth, limiting studies to those prospective in nature ensured that temporality could be established between load and subsequent injury. Finally, use of the NOS, which the Cochrane Handbook cites as one of the best tools to evaluate the methodological quality of non-randomized studies [31], coupled with the fact that most included studies were of good quality on this scale, strengthen the results of this review.

There are also several limitations to this review. We limited our search to publications in the English language, potentially eliminating some high-quality studies from our results.

We also excluded conference proceedings and abstracts, potentially resulting in publication bias since the proportion of studies with null findings is likely greater in those presented at conferences but not published compared with those published in peer-reviewed journals [80, 81]. Second, all study data were extracted by only one author, increasing the possibility of error. Third, the studies evaluated varied widely in their definition of injury, with some including only time-loss injuries diagnosed by a clinician and others including any symptom self-reported by an athlete. This, along with heterogeneity in the types of training load measured, precluded the performance of a meta-analysis. Finally, although the NOS has been shown to have acceptable levels of reliability, this scale is not as reliable or as comprehensive as the D&B, potentially resulting in decreased detection of bias in the examined studies, particularly in the areas of selection of appropriate study design, statistical analysis, and reporting and generalizability of results, all of which are more extensively covered in the D&B [30].

Future research should attempt to improve upon the limitations and gaps present in the current body of research in this area by (1) emphasizing quantification of the relationship between training load and injuries to specific types of tissues; (2) seeking to determine optimal loading for an individual, accounting for important factors such as sport, sex, and previous injury history; and (3) utilizing measures of subjective ITL in tactical athlete populations, particularly using Foster's sRPE method [17] and the ACWR, as neither of these have yet been investigated in these populations. From a design standpoint, future studies should utilize adequate sample sizes to conduct multivariable analyses adjusted for critical covariates such as sex and previous injury history. Adjusting for these and other pertinent covariates was rare in the studies we reviewed. Studies should also use injury rates instead of proportions when possible to adjust for varying exposure between athletes. This is of particular importance in competitions, in which injuries are more frequent and exposure time varies most widely among athletes. It is also important for future studies to record and report latent periods for injuries to improve understanding of the temporal relationship between load and injury risk to help coaching and strength and conditioning staff reduce injury risk through modifying training program design. Relatively few articles examined injury risk with a latent period > 1 week [18, 40, 51–58]. Most only examined injury risk the week each load measurement was taken, or with a short (e.g. 1 week) latent period. Finally, future research should seek to better elucidate the directionality or directionalities of the relationship between different measures of load and injury risk, as included studies reported statistically significant direct, inverse, and U-shaped relationships between load and injury. This could be accomplished in part by the inclusion of latent periods and utilization of non-linear modeling techniques that do not make assumptions about the directionality or linearity of relationships.

5 Conclusions

This systematic review extends the reviews conducted by Jones et al. [19] and Drew and Finch [20], examining the relationship between training load and injury in athletes. It was extended to include populations of tactical athletes and session frequency as a measure of load. Our results demonstrate that there is some evidence for the existence of a load–injury relationship in tactical athlete populations similar to that seen in competitive athletes. Our results also demonstrate that there is conflicting evidence for a relationship between session frequency and injury. Finally, the evidence for a relationship between training load and injury in competitive athletes has been substantially strengthened in the past 2 years and continues to be strongest for sRPE and the ACWR. The shape and direction of the load–injury relationship varies across studies and may depend on the type and timeframe of load examined.

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Compliance with Ethical Standards

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